INCH-POUND

MIL-H-53096(ME)
7 February 1990

MILITARY SPECIFICATION

HOSES AND HOSE ASSEMBLIES, ELASTOMER:

RUBBER, LIQUID FUEL, NONCOLLAPSIBLE, LOW TEMPERATURE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers noncollapsible, synthetic rubber, liquid fuel transfer hoses and hose assemblies for use at all ambient temperatures from -60 to +95 °F.

1.2 Classification. The hose and hose assemblies shall be the following sizes (inside diameter), classes, fitting styles, and fitting material as specified (see 6.2).

Sizes

1 - 1/2 inch
2 - 3/4 inch
3 - 1 inch
4 - 1-1/4 inch
5 - 1-1/2 inch
6 - 2 inches
7 - 2-1/2 inches
8 - 3 inches
9 - 4 inches
10 - 6 inches

Classes

0 - No fitting on either end
1 - Male fitting on one end and female fitting on other end
2 - Male fitting both ends
3 - Female fitting both ends

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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Fittings

Style O - Not applicable (no fitting on either end)
Style A - Banded shank (reattachable)
Style B - Compression (reattachable-threaded)
Style D - Internal expanded (one time threaded)

Materials

Aluminum
Bronze
Brass
Not applicable

1.3 Length. The length of hose or hose assemblies shall be as specified (see 6.2) and shall be specified ±2 percent, exclusive of fittings, and will be in feet with last number expressed in tenths of a foot.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-B-626	- Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
WW-C-440	- Clamps, Hose (Low-Pressure).

MILITARY

MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware: Packaging of.
MIL-C-27487	- Coupling Halves, Quick-Disconnect, Cam-Locking Type.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards.
FED-STD-H28/10	- Screw-Thread Standards for Federal Services Section 10 American National Hose Coupling and Fire Hose Coupling Threads.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawing. The following other Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

ME

13228E1768	- Gasket, Coupling Half, Quick Disconnect: Cam-Locking Type for Arctic Use.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-FS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 176	- Copper-Alloy Castings.
D 380	- Rubber Hose.
D 381	- Existent Gum in Fuels by Jet Evaporation.
D 412	- Rubber Properties in Tension.
D 413	- Rubber Property - Adhesion to Flexible Substrate.
D 471	- Rubber Property - Effect of Liquids.
D 518	- Rubber Deterioration on Surface Cracking.
D 746	- Brittleness Temperature of Plastic and Elastomers by Impact.
D 790	- Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials.
D 1149	- Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen).
D 2137	- Rubber Property - Brittleness Point of Flexible Polymers and Coated Fabrics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hose shall consist of a tube, two or more plies of reinforcement, and a cover. The hose assembly shall consist of one section of hose with each end fitted with couplings, for use with liquid hydrocarbon fuels as specified herein at low temperature environment.

3.1.1 Drawing. The drawing forming a part of this document is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the hose assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Process data. Within the time span specified (see 6.2), the contractor shall make available to the Government, on the site of manufacture, a description of materials and a documented inspection instruction which will be used by the contractor to assure that each production hose assembly is a duplicate of the first article hose assembly. Change of materials, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer.

3.4 Materials. Materials shall be as specified herein and on the drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this document.

3.4.1 Material deterioration prevention and control. The hose assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the hose assembly may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and subcomponents and shall make information available, upon request to the contracting officer or designated representative.

3.4.2. Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the hose assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the hose assembly produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces and parts shall not be incorporated in the hose assembly.

3.4.3 Elastomer.

3.4.3.1 Tube. The basic material for the tube shall be a low temperature elastomer. 1/

3.4.3.2 Cover. The basic material for the cover shall be a low temperature elastomer. 1/

3.4.4 Reinforcement. The hose reinforcement shall be in accordance with 3.4.4.1 or 3.4.4.2, as applicable, and shall have a helix or helixes of round steel wire between the fiber reinforcement; or, hose reinforcement may consist of a carcass of low temperature synthetic filament warp yarns interwoven with a helix or helixes of round, steel wire and fillers of low temperature synthetic filament yarns, alternately spaced. The reinforcing wire shall have the physical and dimensional characteristics required to enable the hose to conform to 3.6.8.

3.4.4.1 Hose up to and including 1-1/2-inches in diameter. The reinforcement shall be two or more plies of low temperature synthetic cord, evenly and firmly fabricated, and thoroughly impregnated with a low temperature elastomeric compound, causing the plies to adhere firmly to each other and to the tube and cover and be of sufficient strength to assure conformance to the applicable requirements of this specification. Dirt, knots, lumps, and irregularity of twist shall be kept at a minimum during fabrication.

3.4.4.2 Hose, 2-inch diameter and larger. The reinforcement may be two or more plies of braided or spiralled low temperature synthetic cord evenly and firmly fabricated and shall be thoroughly impregnated with a low temperature elastomeric compound, causing plies to adhere firmly to each other and to the tube and cover, and be of sufficient strength to conform to the applicable requirements of this specification. As an alternative, the reinforcement may be an even number of layers of parallel nylon woven fabric embedded in low

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1/ NOTE: Suggested materials and procedures for hose fabrication may be found in Report No. 5495410486, published by Goodyear Tire and Rubber Company, 1144 East Market Street, Akron, OH 44316, and Report No. ES-83-C125-1, published by Uniroyal Incorporated, 312 North Hill Street, Mishawaka, IN 46544, and are available under Reports No. AD-B113563L and AD-A139329, respectively, from the Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145.

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temperature elastomeric compound, wound in contradirection, so that there is a rubber bond throughout, providing a continuous rubber bond from the inner tube through the reinforcement to the outer cover. Dirt, knots, lumps, and irregularity of twist shall be kept at a minimum during fabrication.

3.4.5 Fittings (couplings). The fittings (couplings) shall be cast or forged bronze, brass, or aluminum, as specified (see 1.2).

3.4.5.1 Aluminum. Aluminum shall conform to QQ-A-601, alloy 356 temper T-6, or 712.0 temper T-5.

3.4.5.2 Bronze and brass. Bronze shall conform to QQ-C-390, UNS C83600, C85400, C85700, or C92200, type I, in as-cast condition. Brass shall conform to QQ-B-626, alloy 377, in as-forged condition, or ASTM B 176 UNS C87900.

3.4.5.3 Style A, banded shank. Style A fittings shall be cam-locking, quick-disconnect fittings conforming to MIL-C-27487, type II or type VI, class 1 or class 2.

3.4.5.3.1 Clamps. Hose clamps shall conform to WW-C-440, type H, and shall be corrosion-resistant steel. Two clamps shall be used for each fitting, except for 3-inch and larger hose, which shall have three clamps for each fitting. The widths of clamps shall be a minimum 1/2-inch for hose size up to and including 2-1/2-inches inside diameter (ID), and 3/4-inch for hose 3 inches ID and up.

3.4.5.4 Style B. Reattachable compression fittings shall be three-piece construction. No cement, sealers, or adhesives shall be used in their installation other than to seal the cut ends of the hose. The body of the coupling shall be cast or forged with a smooth bore and shall include central hexagon or octagon wrench surfaces and a tailpiece consisting of two concentric walls with an intermediate hose chamber. The outer wall of the tailpiece shall have male threads to engage the sleeve and a tapered end to compress the grip. The inner wall shall be constructed so as to insure a tight grip on the hose and shall have a rounded outer edge to prevent cutting of the hose during installation or upon sharp bending. The grip shall be reusable, left-hand wound into a coil of approximately three loops and may be round or flat with rounded edges. The ends shall be ground square and smooth. The sleeve shall be provided with sturdy lugs offering generous wrench surfaces. The sleeve shall be threaded and tapered in order to facilitate assembly and provide maximum compression of the hose. When the coupling is brass, the sleeve may be bronze conforming to QQ-C-390, copper alloy UNS C85700 or UNS C85400, type I, in as-cast condition, or ASTM B 176, copper alloy UNS C87900. The edge of the sleeve in contact with the hose shall be rounded to prevent cutting of the hose upon sharp bending and shall extend beyond the end of the coupling tailpiece. Body, sleeve, and grip shall be free from porosity and other imperfections.

3.4.5.5 Style D. The one-time-use, internal expanded fittings shall have an internal-expanding serrated tailpiece, suitable gripping, and compressing the wall of the hose against an external ferrule. When expanded, these fittings shall have an inside diameter conforming to the normal inside diameter of the hose.

3.4.6 Threads. Styles B and D fittings shall have tapered male or NPSH female threads. The threads shall conform to FED-STD-H28/10.

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3.5 Construction.

3.5.1 Hose. The hose shall be constructed of a compounded rubber inner tube, synthetic fiber reinforcement with two helically wound reinforcing wires, and a compound rubber cover.

3.5.1.1 Tubing. The tubing shall be fabricated from the material specified in 3.4.3.1. The inside wall of the tube shall be smooth and free from pits. The tube shall be uniform thickness not less than 5/64-inch.

3.5.1.2 Cover. The cover shall be black, fabricated from material specified in 3.4.3.2, and shall be of uniform thickness not less than 5/64-inch.

3.5.1.3 Reinforcement wires. The reinforcing wire (see 3.4.4) shall be continuous to effect an electric bond (see 6.4.1) between both ends of the hose when tested in accordance with 4.5.2.1, and shall withstand the crush resistance requirements in table I when tested in accordance with 4.5.2.9.

TABLE I. Test loads and pressures.

Size of hose (inside diameter) (inches)	Proof test pressure (psi)	Burst test pressure (psi)	Crush resistance (pounds)
1/2	150	500	200
3/4	150	500	200
1	150	500	200
1-1/4	150	500	200
1-1/2	150	500	225
2	125	400	250
2-1/2	125	375	325
3	100	300	325
4	100	300	325
6	100	300	325

3.5.2 Hose assemblies. Hose assemblies shall be hose with fittings attached as specified herein. Elastomeric gaskets, normally supplied with quick-disconnect couplings, shall be replaced with fluorosilicone gaskets conforming to Drawing 13228E1768.

3.5.2.1 Fittings. Fittings shall be securely attached to the hose to form a permanent electric bond with the stiffening wires. The hose assembly shall not leak when tested in accordance with 4.5.2.1.

3.5.3 Diameters and tolerances of hose. Hose diameters and tolerances shall be as shown in table II.

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TABLE II. Hose diameters.

Inside diameter	Outside diameter
1/2 ±1/32 inch	1-1/16 ±1/32 inch
3/4 ±1/32 inch	1-3/8 ±1/32 inch
1 ±1/16 inch	1-5/8 ±1/16 inch
1-1/4 ±1/16 inch	1-7/8 ±1/16 inch
1-1/2 ±1/16 inch	2-1/8 ±1/16 inch
2 ±1/16 inch	2-21/32 ±1/16 inch
2-1/2 ±1/16 inch	3-3/16 ±1/16 inch
3 ±1/16 inch	3-3/4 ±1/16 inch
4 ±1/16 inch	4-3/4 ±1/16 inch
6 ±1/16 inch	6-7/8 +1/8 -1/16 inch

3.6 Physical and chemical requirements.

3.6.1 Hydrostatic pressure.

3.6.1.1 Hose assembly. When tested as specified in 4.5.2.1 at the appropriate proof pressure (see table I), the hose or hose assemblies shall not leak nor show any imperfections in either the hose or the fittings, and there shall be no indication of slippage or pullout of the fitting from the hose. The length of the hose shall not change more than ±7 percent.

3.6.1.2 Minimum burst. When tested as specified in 4.5.2.2, each size hose or hose assembly shall have no failure of hose under the applicable minimum burst pressure shown in table I.

3.6.2 Tensile strength. The tensile strength of the tube and cover shall be not less than 1250 pounds per square inch (psi) when tested as specified in 4.5.2.3. The tensile strength of the tube and cover after immersion, when tested as specified in 4.5.2.4, shall be not less than 40 percent or less than 600 psi (whichever is greater) of the tensile strength obtained before immersion.

3.6.3 Ultimate elongation. When tested as specified in 4.5.2.3, the ultimate elongation of the tube and cover shall be not less than 200 percent, and, when tested as specified in 4.5.2.4, the elongation after immersion shall be not less than 100 percent.

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3.6.4 Adhesion.

3.6.4.1 Original. The original adhesion between tube and plies, between the plies, and between the cover and the plies shall be a minimum load of 10 pounds when tested as specified in 4.5.2.5.

3.6.4.2 After fuel immersion. After fuel immersion, adhesion between the tube and plies, between the plies, and between the cover and the plies shall be a minimum load of 6 pounds after testing as specified in 4.5.2.6.

3.6.5 Volume increase. When tested as specified in 4.5.2.4, the volume increase of the tube specimen, after immersion in test fluid, shall not exceed 60 percent and the volume increase of the cover specimen shall not exceed 100 percent.

3.6.6 Low temperature flexibility.

3.6.6.1 Brittleness. Low temperature brittleness shall be determined for tube and cover material after fuel extraction in accordance with ASTM D 746 at -60 °F. There shall be no evidence of cracking of the tube or cover material when tested in accordance with 4.5.2.7.1.

3.6.6.2 Modulus of elasticity. The modulus of elasticity at -60 °F for the tube and cover materials as determined in accordance with 4.5.2.7.2, shall be less than 1×10^5 psi for the tube and less than $.05 \times 10^5$ psi for the cover.

3.6.7 Existent gum content. When tested as specified in 4.5.2.8, the existent gum residue extracted with heptane shall be not more than 20 milligrams per 100 milliliters of test fluid.

3.6.8 Crush resistance. During the crush resistance test specified in 4.5.2.9, the smallest outside diameter shall be not less than 85 percent of the original outside diameter. After release of the load, the smallest outside diameter shall be not less than 95 percent of the original outside diameter.

3.6.9 Ozone resistance. Ozone resistance shall comply with ASTM D 1149. At the end of the exposure time, there shall be no visible cracking in the cover with 7X magnification when tested in accordance with 4.5.2.10.

3.7 Identification marking. Each length of the hose shall be branded with the contractor's name or trademark, the quarter and year of manufacture, the words "Liquid Petroleum Fuel, Noncollapsible, Low Temperature", and the part or identifying number (see 6.5). The letters shall be molded either recessed or in relief and shall be not less than 1/2-inch high. Hoses shall be branded at 10-foot intervals or by continuous impression branding, provided the pattern will be repeated every 5 feet or less. Label color shall contrast with the background color of the hose.

3.8 Workmanship. The hose and hose assemblies shall conform to the quality specified herein with no cuts, nicks, bruises, blister, sharp edges, or lumps, and shall be clean and smooth.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operatings, is an acceptable practice to ascertain conformance to requirements, however, this quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article hose or hose assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article hose or hose assembly shall be tested as specified in 4.5.2.1 through 4.5.2.10, inclusive. Failure of any test shall be cause for rejection of the hose or hose assembly.

4.4 Quality conformance inspection.

4.4.1 Lot. Each lot shall consist of all units of hose or hose assemblies of the same size, class, style, and composition, manufactured under the same condition and at the same time, and offered to the Government at one time.

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4.4.2 Sampling. Sampling of hose and hose assemblies for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.4 Test.

4.4.4.1 Individual. Each length of hose or each hose assembly shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.4.2 Samples. Three or more units shall be selected from the samples selected in accordance with 4.4.2 and shall be tested as specified in 4.5.2.2 through 4.5.2.10, inclusive. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The hose and hose assembly shall be examined as specified for the following defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
106. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the hose and hose assemblies.
107. Hose not as specified.
108. Hose cut or damaged by fitting assembly.
109. Hose cover scuffed or bruised, so that braid is exposed.
110. Hose cover not clean and smooth.
111. Couplings halves and gaskets not as specified.
112. Clamps not as specified.
113. Identification marking missing, illegible, or not as specified.
114. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Hydrostatic. Each length of hose or hose assembly shall be subjected to the hydrostatic tests of ASTM D 380 for elongation, twist, and proof pressure, using a test fluid that is compatible with the material and operates successfully at -60 °F. The test pressures shall be as specified in table I. The electric bond (see 6.4.1) shall be determined while full test pressure is applied. Nonconformance to 3.5.1.3, 3.5.2.1, or 3.6.1.1 shall constitute failure of this test.

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4.5.2.2 Burst pressure. The hose or hose assembly shall be subjected to the straight bursting test specified in ASTM D 380. Nonconformance to 3.6.1.2 shall constitute failure of this test.

4.5.2.3 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation of the rubber cover and the rubber tube of the hose or hose assemblies shall be determined in accordance with ASTM D 412. Nonconformance to 3.6.2 and 3.6.3 shall constitute failure of these tests.

4.5.2.4 Tensile strength, ultimate elongation, and volume change after fuel immersion. After fuel immersion, the changes in tensile strength, ultimate elongation, and volume change of the rubber cover and rubber tube of hose or hose assemblies selected in accordance with 4.4.2 shall be determined in accordance with ASTM D 471. The temperature of immersion shall be 23 ±1.1 °C. The immersion time shall be 48 hours. The immersion fuel shall be ASTM D 471, reference fuel B. The immediate deteriorated properties method shall be used in determining changes in tensile strength and ultimate elongation. Nonconformance to 3.6.2, 3.6.3, or 3.6.5 shall constitute failure of these tests.

4.5.2.5 Original adhesion. From hose or hose assemblies selected under 4.4.2, prepare ring or strip specimens as described in ASTM D 380. The adhesion shall be determined in accordance with the machine method of ASTM D 413. Nonconformance to 3.6.4.1 shall constitute failure of this test.

4.5.2.6 Adhesion after immersion in fuel. Specimens prepared as described in 4.5.2.5 shall be immersed in fluid as specified in 4.5.2.4 for 48 hours at 23 ±1.1 °C. Within 15 minutes after removal of the specimens from the fluid, the adhesion shall be determined in accordance with machine method of ASTM D 413. Nonconformance to 3.6.4.2 shall constitute failure of this test.

4.5.2.7 Low temperature flexibility.

4.5.2.7.1 Brittleness. Fabricate ASTM slabs of tube and cover material (6 inches X 6 inches X .080-inch) per ASTM D 2137. Condition samples in accordance with ASTM D 471, reference fuel B, at 73 ±2 °F for 4 days. Remove the samples from the fuel and air dry them for 2 days at 73 ±2 °F. Then, oven-age the samples for 24 hours at 130 ±2 °F. Test the samples after oven-aging in accordance with ASTM D 746. Nonconformance to 3.6.6.1 shall constitute failure of this test.

4.5.2.7.2 Modulus of elasticity. Determine the modulus of elasticity at -60 °F per ASTM D 790, method II, procedure B, on samples of tube and cover compounds after they have been conditioned in ASTM D 471, reference fuel B, for 4 days, then air dried and oven-aged as specified in 4.5.2.7.1. Nonconformance to 3.6.6.2 shall constitute failure of this test.

4.5.2.8 Existent gum. A test specimen of not less than 14 inches long shall be cut from the hose selected in accordance with 4.4.2. The bottom end of the specimen shall be stoppered with a clean noncorrosive plug, 2 inches long, secured in the end of the specimen with a suitable clamp. The specimen shall then be filled to within 2 inches from top with fluid as specified in 4.5.2.4 and plugged in a manner similar to the bottom. This specimen shall then be stored in a vertical position for 7 days at an ambient temperature of 38 ±2 °C. At the end of each 24-hour period, the fuel shall be agitated for 5 minutes by moving the

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hose from the vertical to the horizontal position at a rate of 2 cycles per minute. At the completion of the 7-day storage period, fuel shall again be agitated in the specimen for 5 minutes and immediately removed from the hose. The removed fuel shall be tested for existent gum in accordance with the air-jet solvent wash method of ASTM D 381. A sample of test fuel (before entering hose) shall be tested at the same time and by the same method. The existent gum of this sample shall be subtracted from the existent gum obtained from the fuel removed from the test hose. Nonconformance to 3.6.7 shall constitute failure of this test.

4.5.2.9 Crush resistance. Crush resistance shall be determined by centering a 12-inch length of hose between 3-inch wide, parallel metal plates in such a way that a 3-inch length of hose is being compressed. The plates shall be brought together at a rate of 2 inches per minute until the applicable crush resistance load specified in table I has been applied. The distance between plates shall be measured with a steel rule and expressed in percent of the original outside diameter. The load shall be released and the minimum outside diameter of the hose at the center of the compressed area shall be measured and reported in percent of original outside diameter. Nonconformance to 3.5.1.3 or 3.6.8 shall constitute failure of this test.

4.5.2.10 Ozone resistance. Specimens of the cover, prepared as described in procedure A of ASTM D 518, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D 1149. After conditioning for 24 hours in an ozone-free atmosphere, the mounted specimens shall be exposed for 72 hours at 40 ±1 °C, to an atmosphere containing 50 ±10 parts per hundred million (pphm) of ozone. Upon completion of the test, the specimens shall be examined. Nonconformance to 3.6.9 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking of the hoses and hose assemblies shall be examined to determine compliance with the requirements of section 5.

4.7 Inspection comparison. The Government may select hose assemblies at any time during the contract production period and subject these hose assemblies to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article hose assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional test shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any hose assembly to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the hose assembly and shall be cause for refusal by the Government to continue acceptance of production hose assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on hose assemblies produced and accepted under the contract. Any

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deficiencies found as a result of comparison inspection will be considered prima facie evidence that all hose assemblies accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each hose and hose assembly shall be preserved in accordance with the level A preservation requirements of MIL-H-775.

5.1.2 Level C. Each hose and hose assembly shall be preserved in accordance with level C preservation requirements of MIL-H-775.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The hose and hose assemblies shall be packed in accordance with the level A packing requirements of MIL-H-775.

5.2.2 Level B. The hose and hose assemblies shall be packed in accordance with the level B packing requirements of MIL-H-775.

5.2.3 Level C. The hose and hose assemblies shall be packed in accordance with the level C packing requirements of MIL-H-775.

5.3 Marking. Marking for level A, B, and C shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose and hose assemblies covered by this specification are for suction of liquid fuels such as high aromatic gasoline, JP-4, and diesel fuels. The hose assemblies are operable throughout the ambient temperature range of -60 °F (-51 °C) to +95 °F (+32 °C) and will not be damaged by storage throughout the ambient temperature range of +145 °F (+71 °C) to -65 °F (-54 °C).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Size, class, fitting style, and fitting material of hose and hose assembly required (see 1.2).
c. Length of hose or hose assembly required (see 1.3).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. Part or identifying number required (see 3.7 and 6.5).
f. When a first article is not required (see 3.2) and the time frame specified for process data (see 3.3).
g. Level of preservation and level of packing required (see 5.1 and 5.2).

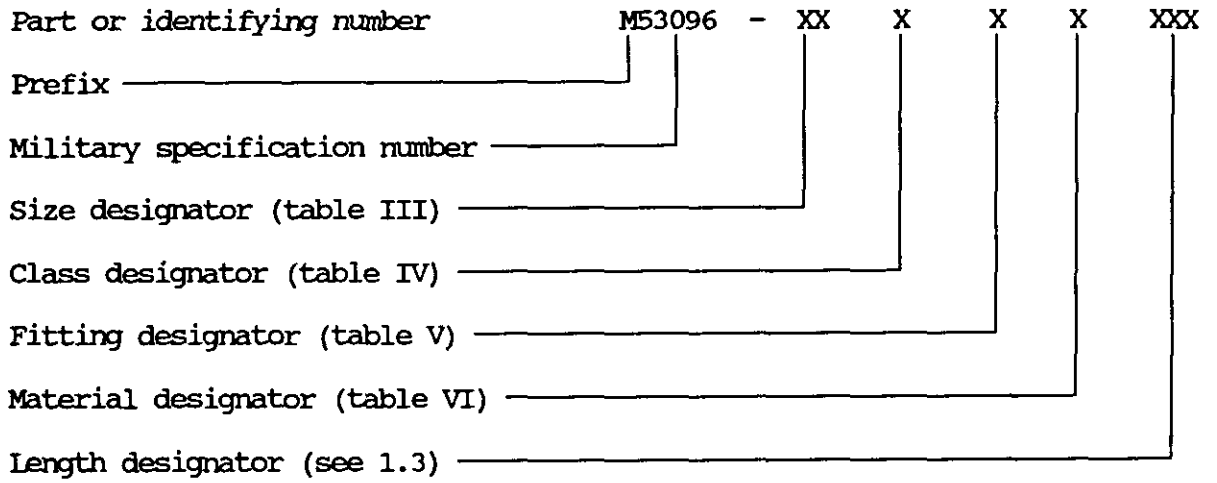
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6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Electrical bond. An electrical bond eliminates a difference in potential between objects.

6.5 Part or identifying number (PIN). The PIN to be used for hose assemblies acquired to this specification are created as follows (see 3.7 and 6.2).



Example 1: M53096-01B2A500 - Hose assembly in accordance with MIL-H-53096, size 1 (1/2 inch), class 1 (male fitting on one end and female fitting on other end), style A (banded shank - reattachable), material - aluminum, length 50.0 feet.

Example 2: M53096-01A1D500 - Hose in accordance with MIL-H-53096, size 1 (1/2 inch), class 0 (no fitting on either end), style 0 (not applicable - no fitting on either end), material - not applicable (no fitting on either end), length 50.0 feet.

6.5.1 Size. The hose and hose assembly size are identified by a two-digit code number as shown in table III.

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TABLE III. Size designator.

Code No.	Inside Diameter
01	Size 1 - 1/2 inch
02	Size 2 - 3/4 inch
03	Size 3 - 1 inch
04	Size 4 - 1-1/4 inches
05	Size 5 - 1-1/2 inches
06	Size 6 - 2 inches
07	Size 7 - 2-1/2 inches
08	Size 8 - 3 inches
09	Size 9 - 4 inches
10	Size 10 - 6 inches

6.5.2 Class. The hose assembly class is identified by a one-letter symbol as shown in table IV.

TABLE IV. Class designator.

Symbol	Characteristic
A	Class 0 - No fitting on either end
B	Class 1 - Male fitting on one end and female fitting on other end
C	Class 2 - Male fitting both ends
D	Class 3 - Female fittings both ends

6.5.3 Fitting. The hose assembly fitting style is identified as a single code numeral as shown in table V.

TABLE V. Fitting designator.

Code No.	Characteristic
1	Style 0 - Not applicable
2	Style A - Banded shank (reattachable)
3	Style B - Compression (reattachable-threaded)
4	Style D - Internal expanded (one time threaded)

6.5.4 Material. The hose assembly fitting material is identified by a one-letter symbol as shown in table VI.

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TABLE VI. Material designator.

Symbol	Metal
A	Aluminum
B	Bronze
C	Brass
D	Not applicable

6.6 Subject term (key word) listing.

Aluminum
Arctic
Brass
Bronze
Clamps
Coupling
Cover
Diesel fuel
Fittings
Fuel
Fuels, liquid hydrocarbon
Gasoline, high aromatic
Gasoline
Hose
JP-4
Tube

Custodian:
Army - ME

Preparing activity:
Army - ME

(Project 4720-A038)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-53096(ME)	900207

3. DOCUMENT TITLE: Hoses and Hose Assemblies, Elastomer: Rubber, Liquid Fuel, Noncollapsible Low Temperature

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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